

LiFi Transmission Code(For UNO)

```
#include<SoftwareSerial.h>
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"
#include "DHT.h"

#include "I2Cdev.h"
#include "MPU6050.h"

MPU6050 mpu;

#define DHTPIN 7
#define DHTTYPE DHT11
#define REPORTING_PERIOD_MS      1000
PulseOximeter pox;

uint32_t tsLastReport = 0;
String data;
float Hr,SpO2,t;

int16_t ax, ay, az;
int16_t gx, gy, gz;
SoftwareSerial mySerial(2,3);
DHT dht(DHTPIN, DHTTYPE);
void onBeatDetected()
{
    Serial.println("Beat!");
}

void setup()
{
    Serial.begin(9600);
    mySerial.begin(9600);
    dht.begin();
    Wire.begin();
    mpu.initialize();
    Serial.print("Initializing pulse oximeter..");
    if (!pox.begin()) {
        Serial.println("FAILED");
        for(;;);
    } else {
        Serial.println("SUCCESS");
    }
}
```

```

    pox.setOnBeatDetectedCallback(onBeatDetected);
    mpu.setXAccelOffset(0); //Set your accelerometer offset for axis X
    mpu.setYAccelOffset(0); //Set your accelerometer offset for axis Y
    mpu.setZAccelOffset(0); //Set your accelerometer offset for axis Z
    mpu.setXGyroOffset(0); //Set your gyro offset for axis X
    mpu.setYGyroOffset(0); //Set your gyro offset for axis Y
    mpu.setZGyroOffset(0);

}

void loop()
{

    pox.update();
    mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);
    if (millis() - tsLastReport > REPORTING_PERIOD_MS) {
        Serial.print("Heart rate:");
        Serial.print(pox.getHeartRate());
        Serial.print("bpm / SpO2:");
        Serial.print(pox.getSpO2());
        t = dht.readTemperature(true);
        Hr=pox.getHeartRate();
        SpO2=pox.getSpO2();
        Serial.println("%");
        data=String(Hr)+", "+String(SpO2)+", "+String(t)+", "+String(ax);
        Serial.println(data);
        mySerial.println(data);
        tsLastReport = millis();

    }
}

```

LiFi Receiver Code(For ESP32)

```
#define BLYNK_TEMPLATE_ID "TMPL3BFqj0tTW"
#define BLYNK_TEMPLATE_NAME "LIFI"
#define BLYNK_AUTH_TOKEN "zET0piORCCrWLyAP2pF6055dkadKM9Y8"

#include<LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,16,2);

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>

WiFiClient client;
char auth[] = "zET0piORCCrWLyAP2pF6055dkadKM9Y8";
char ssid[] = "iotdata";
char pass[] = "12345678";

#include<SoftwareSerial.h>
SoftwareSerial mySerial(14, 27);
int temp,Hr,ax,SpO2;
String dat;
void setup() {
    Serial.begin(9600);
    mySerial.begin(9600);
    lcd.init();
    lcd.init();
    lcd.backlight();
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("wifi connecting.....");
    Blynk.begin(auth,ssid,pass);
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("wifi connected");
    delay(300);
    lcd.clear();
    lcd.print("LIFI RECEIVER");
    delay(1000);
}

void loop() {
    Blynk.run();
    if(mySerial.available()>0){
```

```

dat = mySerial.readStringUntil('\n');
Serial.print("dat = ");
Hr=dat.substring(0,dat.indexOf(',')).toInt();
dat.remove(0,dat.indexOf(',')+1);
SpO2=dat.substring(0,dat.indexOf(',')).toInt();
dat.remove(0,dat.indexOf(',')+1);
temp=dat.substring(0,dat.indexOf(',')).toInt();
dat.remove(0,dat.indexOf(',')+1);
ax=dat.substring(0,dat.indexOf(',')).toInt();
dat.remove(0,dat.indexOf(',')+1);
}

Serial.print("Hr= ");
Serial.println(Hr);
Serial.print("SpO2 =");
Serial.println(SpO2);
Serial.print("temp= ");
Serial.println(temp);
Serial.print("ax= ");
Serial.println(ax);
Serial.print("dat =");
Serial.println(dat);

Blynk.virtualWrite(V0,Hr);
Blynk.virtualWrite(V1,SpO2);
Blynk.virtualWrite(V2,temp);
Blynk.virtualWrite(V3,ax);

if(Hr>120){
  Blynk.logEvent("Hr","HIGH HEART RATE ALERT");
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("HIGH HEART");
  lcd.setCursor(0,1);
  lcd.print("ALERT");
  delay(500);
}
else if(temp>100){
  Blynk.logEvent("temp","HIGH TEMPERATRE ALERT");
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("HIGH TEMPERATURE");
  lcd.setCursor(0,1);
  lcd.print("ALERT");
  delay(500);
}
else{

```

```
    dis();  
  }  
  Blynk.virtualWrite(V0,Hr);  
  Blynk.virtualWrite(V2,temp);  
  Blynk.virtualWrite(V3,ax);  
  Blynk.virtualWrite(V1,SpO2);  
  
}  
  
void dis(){  
  lcd.clear();  
  lcd.setCursor(0,0);  
  lcd.print("HR = ");  
  lcd.print(Hr);  
  lcd.setCursor(0,1);  
  lcd.print("SpO2 = ");  
  lcd.print(SpO2);  
  delay(500);  
  lcd.setCursor(0,0);  
  lcd.print("Temp = ");  
  lcd.print(temp);  
  lcd.setCursor(0,1);  
  lcd.print("Ax = ");  
  lcd.print(ax);  
  delay(500);  
}
```